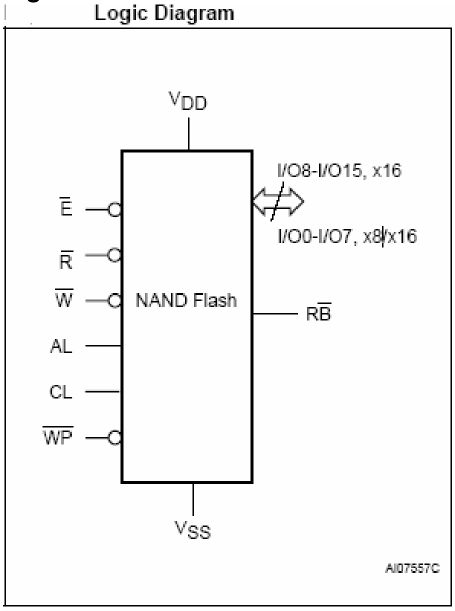
NAND512W3A

512 Gbit
528 Byte/264 Word Page, 3.0V, Supply Flash Memory
Vhdl Behavioral Model Documentation

FEATURES SUMMARY

- HIGH DENSITY NAND FLASH MEMORIES
  – Up to 1 Gbit memory array
  – Up to 32 Mbit spare area
  – Cost effective solutions for mass storage applications
- NAND INTERFACE
  – x8 or x16 bus width
  – Multiplexed Address/ Data
  – Pinout compatibility for all densities
- SUPPLY VOLTAGE
  – 1.8V device: V_{DD} = 1.7 to 1.95V
  – 3.0V device: V_{DD} = 2.7 to 3.6V
- PAGE SIZE
  – x8 device: (512 + 16 spare) Bytes
  – x16 device: (256 + 8 spare) Words
- BLOCK SIZE
  – x8 device: (16K + 512 spare) Bytes
  – x16 device: (8K + 256 spare) Words
- PAGE READ / PROGRAM
  – Random access: 12µs (max)
  – Sequential access: 50ns (min)
  – Page program time: 200µs (typ)
- COPY BACK PROGRAM MODE
  – Fast page copy without external buffering
- FAST BLOCK ERASE
  – Block erase time: 2ms (Typ)
- STATUS REGISTER
- ELECTRONIC SIGNATURE
- CHIP ENABLE ‘DON’T CARE’ OPTION
  – Simple interface with microcontroller
- SERIAL NUMBER OPTION
- HARDWARE DATA PROTECTION
  – Program/Erase locked during Power transitions

Figure 1.
Logic Diagram



- DATA INTEGRITY
  – 100,000 Program/Erase cycles
  – 10 years Data Retention
- RoHS COMPLIANCE
  – Lead-Free Components are Compliant with the RoHS Directive
- DEVELOPMENT TOOLS
  – Error Correction Code software and hardware models
  – Bad Blocks Management and Wear Leveling algorithms
  – File System OS Native reference software
  – Hardware simulation models

TABLE OF CONTENTS

INTRODUCTION

In order to describe the *Model Documentation,* in this document you can find useful informations for the device in object, its features and characteristics, jointly the structure of the delivered model and how to use it.
The *Model Documentation* consists of two sections:

- The *Device Description* describes the main device features and its characteristics, and allows the prompt understanding of the most important functionality of the device.
  It is from the device target specification datasheet.

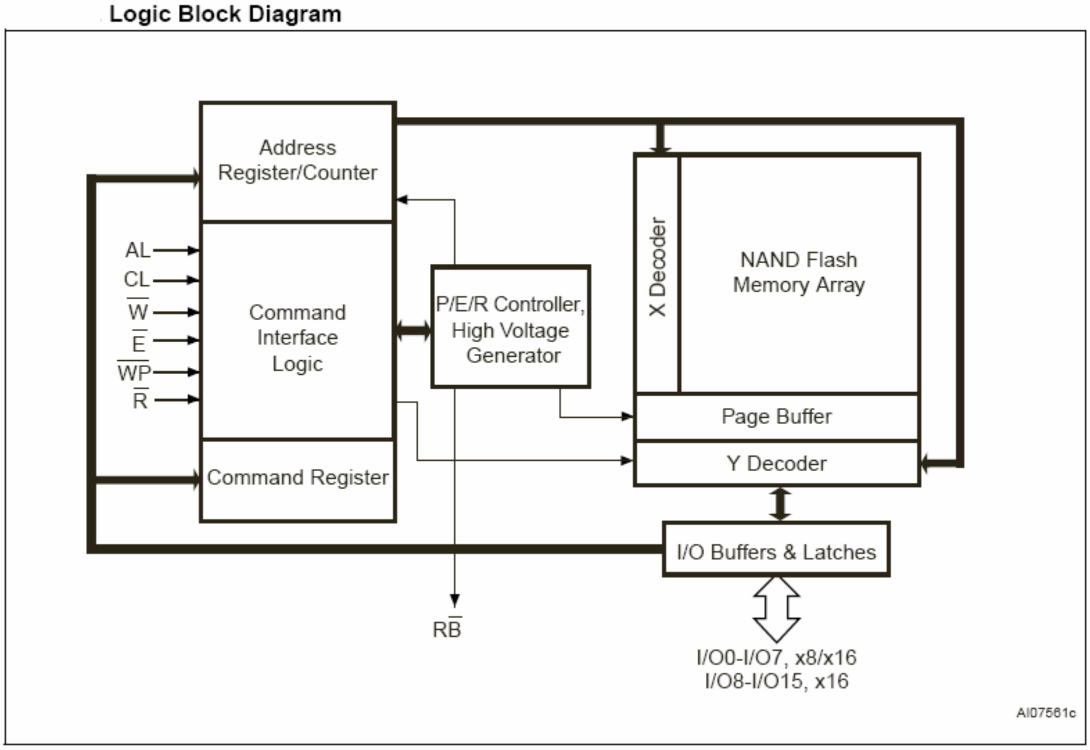
- The section *Application Note* describes how the delivered model was thought and implemented, and also all steps that you have to perform to use.
  It concerns the simulation process, explains the design decisions and the structure of the model.

SECTION 1 – DEVICE DESCRIPTION

SUMMARY DESCRIPTION

The NAND Flash 528 Byte/ 264 Word Page is a family of non-volatile Flash memories that uses the Single Level Cell (SLC) NAND cell technology.
It is referred to as the Small Page family. The devices range from 128Mbits to 1Gbit and operate with either a 3V voltage supply. The size of a Page is either 528 Bytes (512 + 16 spare) or 264 Words (256 + 8 spare) depending on whether the device has a x8 or x16 bus width.The address lines are multiplexed with the Data Input/Output signals on a multiplexed x8 or x16 Input/Output bus. This interface reduces the pin count and makes it possible to migrate to other densities without changing the footprint.
Each block can be programmed and erased over 100,000 cycles. To extend the lifetime of NAND Flash devices it is strongly recommended to implement an Error Correction Code (ECC). A Write Protect pin is available to give a hardware protection against program and erase operations.
The devices feature an open-drain Ready/Busy output that can be used to identify if the Program/Erase/Read (P/E/R) Controller is currently active.
The use of an open-drain output allows the Ready/Busy pins from several memories to be connected to a single pull-up resistor.
A Copy Back command is available to optimize the management of defective blocks. When a Page Program operation fails, the data can be programmed in another page without having to resend the data to be programmed.

Logic Block Diagram



SIGNAL DESCRIPTION

Data Inputs/Outputs (I/O0-7).
The Data I/O are used to select the cells to access in the memory array, output the data stored at the selected address or input a command or a data to be programmed for x8 or x16 devices .

Data Inputs/Output (I/O8-15).
Data Input/Output x16 devices.

Chip Enable (E).
The Chip Enable input activates the memory control logic. When the Chip Enable is Low the device is in active mode.

Read Enable (R).
The Readle input activates the Bus Read Operations when it is Low.

Write Enable (W).
The Write Enable input activates the Bus Write Operations when it is Low.

Write Protect (WP).
Write Protect input gives an additional hardware protection for each block. When it is Low, Locked-Down is enabled.

Ready Busy (RB).
Ready Busy (open drain input) that can be used to identify if the P/E/R Controller is active.

Address Latch Enable (AL).
Address Latch Enable latches the address bits on its rising edge.

Command Latch Enable (CL).
Command Latch Enable activates the latching of the Command Input.

Signal Names

I/O8-15	Data Input/Outputs for x16 devices
I/O0-7	Data Input/Outputs, Address Inputs, or Command Inputs for x8 and x16 devices
AL	Address Latch Enable
CL	Command Latch Enable
$\overline{E}$	Chip Enable
$\overline{R}$	Read Enable
R$\overline{B}$	Ready/Busy (open-drain output)
$\overline{W}$	Write Enable
$\overline{WP}$	Write Protect
V_{DD}	Supply Voltage
V_{SS}	Ground
NC	Not Connected Internally
DU	Do Not Use

Table 1. Signals.

V_{DD} Supply Voltage.
V_{DD} provides the power supply to the internal core of the memory device. It is the main power supply for all operations (Read, Program and Erase).

V_{SS} Ground.
V_{SS} ground, is the reference for the core power supply. It must be connected to the system ground.

NC.
Not Connected Internally.

DU.
Do not use

BUS OPERATIONS

There are six standard bus operations that control the device. These are Command Input, Address Input, Data Input, Data Output, Write Protect and Standby.

Command Input.
Command Input bus operations are used to give commands to the memory. Commands are accepted when Chip Enable is Low, Command Latch Enable is High, Address Latch Enable is Low and Read Enable is High. They are latched on the rising edge of the Write Enable signal. Only I/O0 to I/O7 are used to input commands.

Address Input.
Address Input bus operations are used to input the memory addresses. Four bus cycles are required to input the addresses for the 512Mb and 1Gb devices

Data Input.
Data Input bus operations are used to input the data to be programmed.
Data is accepted only when Chip Enable is Low, Address Latch Enable is Low, Command Latch Enable is Low and Read Enable is High. The data is latched on the rising edge of the Write Enable signal. The data is input sequentially using the Write Enable signal.

Data Output.
Data Output bus operations are used to read: the data in the memory array, the Status Register, the lock status, the Electronic Signature and the Unique Identifier.Data is output when Chip Enable is Low, Write Enable is High, Address Latch Enable is Low, and Command Latch Enable is Low. The data is output sequentially using the Read Enable signal.

Write Protect.
Write Protect bus operations are used to protect the memory against program or erase operations.
When the Write Protect signal is Low the device will not accept program or erase operations and so the contents of the memory array cannot be altered.
The Write Protect signal is not latched by Write Enable to ensure protection even during power-up.

Standby.
When Chip Enable is High the memory enters Standby mode, the device is deselected, outputs are disabled and power consumption is reduced.

Table 2. Bus Operation

Bus Operation	E	AL	CL	R	W	WP	I/O0 - I/O7	I/O8 - I/O15(1)
Command Input	V_{IL}	V_{IL}	V_{IH}	V_{IH}	Rising	X(2)	Command	X
Address Input	V_{IL}	V_{IH}	V_{IL}	V_{IH}	Rising	X	Address	X
Data Input	V_{IL}	V_{IL}	V_{IL}	V_{IH}	V_{IH}	V_{IH}	Data Input	Data Input
Data Output	V_{IL}	V_{IL}	V_{IL}	Falling	X	X	Data Output	Data Output
Write Protect	X	X	X	X	X	V_{IL}	X	X
Standby	V_{IH}	X	X	X	X	X	X	X

Note: 1. Only for x16 devices.
2. WP must be V_{IH} when issuing a program or erase command.

Table 3. Address Insertion x8 device

Bus Cycle	I/07	I/06	I/05	I/04	I/03	I/02	I/01	I/00
1 st	A7	A6	A5	A4	A3	A2	A1	A0
2 nd	A16	A15	A14	A13	A12	A11	A10	A9
3 nd	A24	A23	A22	A21	A20	A19	A18	A17
4 nd(4)	VIL	VIL	VIL	VIL	VIL	VIL	A26	A25

Table 4. Address Insertion x16 device

Bus Cycle	I/07	I/06	I/05	I/04	I/03	I/02	I/01	I/00
1 st	A7	A6	A5	A4	A3	A2	A1	A0
2 nd	A16	A15	A14	A13	A12	A11	A10	A9
3 nd	A24	A23	A22	A21	A20	A19	A18	A17
4 nd	VIL	VIL	VIL	VIL	VIL	VIL	A26	A25

Note: 1. A8 is Don't Care in x16 devices.
2. Any additional address input cycles will be ignored.
3. The 01h Command is not used in x16 devices.
4. The 4th cycle is only required for 512Mb and 1Gb devices.

Table 5. Address Definitions

Address	Definition
A0 - A7	Column Address
A9 - A26	Page Address
A9 - A13	Address in Block
A14 - A26	Block Address
A8	A8 is set Low or High by the 00h or 01h Command, and is Don't Care in x16 devices

COMMAND INTERFACE

All Bus Write Operations are interpreted by the Command Interface. Commands consist of one or more sequential Bus Write Operations.
Each code sequence identifies a unique command.

When the command has been recognized, the Program/Erase controller takes the operative control, handles all timings and verifies the correct command execution.

In the next table you can see all command codes :

Table 7. Command Codes

Command	**Bus Write Operations**(1)			**Command accepted during busy**
	1st CYCLE	**2nd CYCLE**	**3rd CYCLE**	
Read A	00h			
Read B	01h(2)			
Read C	50h			
ReadElectronicSignature	90h			
Read Status Register	70h			Yes
Page Program	80h	10h		
Copy Back Program	00h	8Ah	10h	
Block Erase	60h	D0h		
Reset	FFh			Yes

Note: 1. The bus cycles are only shown for issuing the codes. The cycles required to input the addresses or input/output data are not shown.
2. Any undefined command sequence will be ignored by the device.

STATUS REGISTER

The device contains a Status Register which provides information on the current or previous Program or Erase operation. The various bits in the Status Register convey information and errors on the operation. The Status Register is read by issuing the Read Status Register command. The Status Register information is present on the output data bus (I/O0-I/O7) on the falling edge of Chip Enable or Read Enable, whichever occurs last. When several memories are connected in a system, the use of Chip Enable and Read Enable signals allows the system to poll each device separately, even when the Ready/Busy pins are common-wired. It is not necessary to toggle the Chip Enable or Read Enable signals to update the contents of the Status Register. After the Read Status Register command has been issued, the device remains in Read Status Register mode until another command is issued. Therefore if a Read Status Register command is issued during a Random Read cycle a new read command must be issued to continue with a Page Read or Sequential Row Read operation.The Status Register bits are summarized in Table 8., Status Register Bits. Refer to Table 8. in conjunction with the following text descriptions.

Write Protection Bit (SR7). The Write Protection bit can be used to identify if the device is protected or not. If the Write Protection bit is set to ‘1’ the device is not protected and program or erase operations are allowed. If the Write Protection bit is set to ‘0’ the device is protected and program or erase operations are not allowed.

P/E/R Controller Bit (SR6). The Program/Erase/Read Controller bit indicates whether the P/E/R Controller is active or inactive. When the P/E/R Controller bit is set to ‘0’, the P/E/R Controller is active (device is busy); when the bit is set to ‘1’, the P/E/R Controller is inactive (device is ready).

Error Bit (SR0). The Error bit is used to identify if any errors have been detected by the P/E/R Controller. The Error Bit is set to ’1’ when a program or erase operation has failed to write the correct data to the memory. If the Error Bit is set to ‘0’ the operation has completed successfully.

SR5, SR4, SR3, SR2 and SR1 are Reserved.

Table 8. Status Register Bits

Bit	Name	Logic Level	Definition
SR7	Write Protection	'1'	Not Protected
		'0'	Protected
SR6	Program/ Erase/ Read Controller	'1'	P/E/R C inactive, device ready
		'0'	P/E/R C active, device busy
SR5, SR4, SR3, SR2, SR1	Reserved	Don’t Care	
SR0	Generic Error	'1'	Error – operation failed
		'0	No Error – operation successful

READ MODES

Each operation to read the memory area starts ith a pointer operation as shown in the PointerOperations section. Once the area (main or spare) has been selected using the Read A, Read B or Read C commands four bus cycles (for 512Mb and 1Gb devices) are required to input the address (refer to Table 3.) of the data to be read. The device defaults to Read A mode after powerup or a Reset operation.

When reading the spare area addresses:

- A0 to A3 (x8 devices)
- A0 to A2 (x16 devices)

are used to set the start address of the spare area while addresses:

- A4 to A7 (x8 devices)
- A3 to A7 (x16 devices)

are ignored. Once the Read A or Read C commands have been issued they do not need to be reissued for subsequent read operations as the pointer remains in the respective area. However, the Read B command is effective for only one operation, once an operation has been executed in Area B the pointer returns automatically to Area A and so another Read B command is required to start another read operation in Area B. Once a read command is issued three types of operations are available: Random Read, Page Read and Sequential Row Read.

Random Read. Each time the command is issued the first read is Random Read.

Page Read. After the Random Read access the page data is transferred to the Page Buffer in a time of t_{WHBH} Once the transfer is complete the Ready/Busy signal goes High. The data can then be read out sequentially (from selected column address to last column address) by pulsing the Read Enable signal.

Sequential Row Read. After the data in last column of the page is output, if the Read Enable signal is pulsed and Chip Enable remains Low then the next page is automatically loaded into the Page Buffer and the read operation continues. A Sequential Row Read operation can only be used to read within a block. If the block changes a new read command must be issued. To terminate a Sequential Row Read operation set the Chip Enable signal to High for more than t_{EHEL}.Sequential Row Read is not available when the Chip Enable Don't Care option is enabled.

PROGRAM MODES

PAGE PROGRAM

The Page Program operation is the standard operation to program data to the memory array. The main area of the memory array is programmed by page, however partial page programming is allowed where any number of bytes (1 to 528) or words (1 to 264) can be programmed. The maximum number of consecutive partial page program operations allowed in the same page is three. After exceeding this a Block Erase command must be issued before any further program operations can take place in that page. Before starting a Page Program operation a Pointer operation can be performed to point to the area to be programmed.Each Page Program operation consists of five steps:

1. one bus cycle is required to setup the Page Program command
2. four bus cycles are then required to input the program address (refer to Table 3.) 3. the data is then input (up to 528 Bytes/ 264 Words) and loaded into the Page Buffer
4. one bus cycle is required to issue the confirm command to start the P/E/R Controller.
5. The P/E/R Controller then programs the data into the array. Once the program operation has started the Status Register can be read using the Read Status Register command. During program operations the Status Register will only flag errors for bits set to '1' that have not been successfully programmed to '0'. During the program operation, only the Read Status Register and Reset commands will be accepted, all other commands will be ignored. Once the program operation has completed the P/ E/R Controller bit SR6 is set to '1' and the Ready/ Busy signal goes High.

The device remains in Read Status Register mode until another valid command is written to the Command Interface.

COPY BACK PROGRAM

The Copy Back Program operation is used to copy the data stored in one page and reprogram it in another page.The Copy Back Program operation does not require external memory and so the operation is faster and more efficient because the reading and loading cycles are not required. The operation is particularly useful when a portion of a block is updated and the rest of the block needs to be copied to the newly assigned block.

If the Copy Back Program operation fails an error is signalled in the Status Register. However as the standard external ECC cannot be used with the Copy Back operation bit error due to charge loss cannot be detected. For this reason it is recommended to limit the number of Copy Back operations on the same data and or to improve the performance of the ECC.

The Copy Back Program operation requires three steps:

1. The source page must be read using the ReadA command (one bus write cycle to setup the command and then 4 bus write cycles to input the source page address). This operation copies all 264 Words/ 528 Bytes from the page into the Page Buffer.
2. When the device returns to the ready state (Ready/Busy High), the second bus write cycle of the command is given with the 4 bus cycles to input the target page address.
3. Then the confirm command is issued to start the P/E/R Controller.

After a Copy Back Program operation, a partialpage program is not allowed in the target page until the block has been erased

SECTION 2 – APPLICATION NOTE

VHDL MODEL DELIVERY PACKAGE ORGANIZATION

The Vhdl Model Delivery Package *ST_* NAND512W3A_V1.0*.tar.gz* is organized as hereafter described. A main directory named *ST_*NAND512W3A_*V1.0* contains five subdirectories, with related files.

code (source code files)

- NAND512W3A.vhd

lib (source library files)

- lib/BlockLib.vhd
- ib/CUIcommandData.vhd
- lib /MemoryLib.vhd
- lib/StringLib.vhd
- lib/TimingData.vhd
- lib/UserData.vhd
- lib/data.vhd
- lib/def.vhd

doc (model documentation)

- doc/NANDXXX-A.pdf
- doc/ApplicationNote.pdf

sim (files for simulation)

- sim/memory_file
- sim/run_ncsim

stim (stimuli and testbench files for simulation)

- read/latch.vhd :latch operation, command code 00h Read A
- read/read-A.vhd :read operation, command code 00h ReadA
- read/read-B.vhd : read operation, command code 01h ReadB
- read/read-C.vhd : read operation, command code 50h ReadC
- read/read-err.vhd : read operation, command code error 02h Read not definied
- read/read_signature.vhd: read signature command 90h, address 00h, read cycle
- read/read_status_reg.vhd: read status register, read set up code 00h, read operation
- read/reset.vhd: reset operation, command code FFh
- read/seq_row-areaA-2block.vhd: erase and program 2 page neighboring in 2 different blocks in area A,read pages, sequential read area A
- read/seq_row-areaABC.vhd: sequential read area A, B, C
- read/seq_row-areaC-2block.vhd: erase and program 2 page neighboring in 2 different blocks in area C,read pages, sequential read area C
- read/seq_row-programA.vhd: program 2 page neighboring in the same block, sequential read in area A, B, C

- read/seq_row-programB.vhd: program 2 page neighboring in the same block, in area B, sequential read in area A, B, C
- read/seq_row-programC.vhd: rogram 2 page neighboring in the same block, in area C, sequential read in area C
- program/copy_back-adv.vhd: page program in area A and copy back command
- program/copy_back-nodata.vhd: page program in area A and copy back command without data in target address
- program/copy_back.vhd: page program in area A and copy back command
- program/double-pageprogram-areaA.vhd: double page program in area A and read
- program/double-pageprogram-areaB.vhd: double page program in area B and read
- program/double-pageprogram-areaC.vhd: double page program in area C and read
- program/page_erase-areaA.vhd: page program in area A, block erase and read
- program/page_program-areaA.vhd: page program area A and read area A, B, C
- program/page_program-areaB.vhd: page program area B and read area A, B, C
- program/page_program-areaC.vhd: page program area C and read A, B and C
- program/page_program_err.vhd: page program with read code error
- program/page_read-areaA.vhd: page program and read in area A
- program/page_read-areaABC.vhd: page program in area B and read in area A, B and C
- program/program-areaAB.vhd: double page program area A, page program area B and read
- erase/block_erase.vhd: block erase command code
- erase/block_erase_err.vhd: block erase generates an error
- erase/page_program-erase_3block.vhd: read area A & B, page program area A&B, block erase & read
- erase/page_program-erase_areaAB.vhd: read area A & B, page program area A&B, block erase & read A, B, C

NAND512W3A VHDL BEHAVIOURAL MODEL

The NAND512W3A (V1.0) Vhdl memory model is based on the target specification datasheet December 2004 of 528 Byte/264 Word Page, 3.0V Supply Flash Memory. It contains a lot of modules to implement the functionality of the device.

Modules and Libraries

The Vhdl model of NAND512W3A device consists of modules and libraries. The modules, we have used, are the following :

- **TimingLibModule** checks for all timing constraints;
- **SignatureModule** implements the Electronic Signature Command of the CUI;
- **CUIdecoder** decodes all cycles commands;
- **EraseModule** implements the Block Erase Command of the CUI;
- **MemoryModule** manages all operations on the memory array;

- **OutputBufferModule** manages all operations on the Data Bus;
- **ProgramModule** i implements the Program Command of the CUI;
- **ReadModule** implements the Read Memory Array Command;
- **StatusRegModule** manages the Status Register and implements the Read Status Register Command;
- **KernelModule**, is the kernel of the model;

The libraries contains constants definitions and utility functions and tasks, used in the modules. They are the following :

- ***BlockLib.h***: contains general procedures and functions for representing and managing the banks and the blocks of the device.
- ***StringLib*.h** contains utilities for managing strings.
- ***Memory_lib*.h** contains general procedures for managing the memory array.
- ***CUIcommandData.h***: contains commands code and states of Command User Interface (CUI).
- ***TimingData.h***: contains the definition of constants related with the timing constraints of the device.
- ***UserData.h***: contains the definition of constants that can be changed by users like bank organization (top or bottom), access time (60 ns, 70 ns or 80 ns), TimingChecks (on or off). This is the only library that can be modified by the user, and it will be described in the next section, *Customize your simulation.*
- ***data.h***: contains the definition of constants, types and procedures related with memory characteristics,
- ***def.h***: contains the definition of constants of types.

Both modules and libraries must be compiled in the same order as they were written up.

In the directory **sim**, there is a script to launch the simulation (see run_ncsim).

Customize your simulation.

The simulation process is customizable by the user in some features and characteristics.
The parameters that the user can modify are contained in the ***UserData*** library, in the include directory.

The parameters whose value can be changed are:

- **TimingChecks**; if this parameter is setted to *on* value, no timing checks will be done during the simulation. Else, if it is setted to *off* value, all timing checks will be done during the simulation.
  default value : « **on** »

Vhdl types used in model ports

The port section of NAND512W3A model, defines the name and the related type for each signal of the device.

In particular the following ports are defined:

Port	Type	Description
I/O	standard logic (15 downto 0)	I/O Input /Output
E_N	standard logic	Chip Enable
R_N	standard logic	Output Enable
W_N	standard logic	Write Enable
AL	standard logic	Address Latch Enable
WP_N	standard logic	Write Protect
CL	standard logic	Command Latch Enable
RB	standard logic	Ready/Busy
Vdd	real	Supply Voltage
Vss	real	Ground

Table 10 . Model Ports

Simulation Timings

User will not wait for long time to obtain the result of the simulation. The final result is important, so, in order to reduce some simulation time, the value of some latency times have been reduced.

	Real Time	Simulation Time	Unit
Block Erase Time	2 ms	20 µs	ms
Page Program Time	200	3	µs
Reset Time	5	5	µs

Table 11. Simulation Timings

These values can be fixed by setting the following variables, in the *TimingData.h* library file:

- Erase_time
- Program_time
- Reset_time

Test Data Load File Format

It is possible, for simulation purposes, to preload the memory array with a specific content.
The format of the memory file must be as following:

hex_address/hex_data

For example:

07FFFF/7FFF

The user must write the file name into the entity file:

generic(memoryfile: string := path/filename)

This filename hasn't to be empty.
If the user doesn't provide the initialization file, the memory array is loaded with a sequence word of "1", therefore the matrix is completed erased.

An example of the memory file is the file 'memory_file', in the directory 'sim'.

Version History

February 25, 2005 – Version 1.0

NAND512W3A Vhdl Testbench and Stimuli files

In directory ‘top’ NAND512W3A delivery package, you can find a testbench file (Top.vhd) and in directory ‘stim’ some stimuli files in Vhdl format.

The stimuli files cover a lot of operational conditions of the device, and in particular, the CUI commands.

The testbench and the stimuli files has been written using Vhdl standard version.
Cadence NC-SIM 5.3 simulator has been used to validate the model.
The use of the model with other simulators is not guaranteed.

Authors and Contacts

Both model versions and stimuli files have been developed internally by ST MPG-CAD group. For any issue please contact:

Authors:

Name: Giampaolo Giacopelli
Phone: +39.091.668.9948
e-mail: giampaolo.giacopelli@st.com

Name: Antonella Cavadi
Phone: +39.091.668.9946
e-mail: antonella.cavadi@st.com